

Some initial results of Building a Database of Logic Circuit Graphs for Developing AI Applications in Logic Circuit Analysis & Design

1

Prof.Dr. Do Phuc
University of Information Technology, VNUHCM
Year 2025
Release 11.0



Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

1

2

Outlines

- Use AI to Convert The Logic Circuit to Graph and Build Database of logic Circuit
- Generate Truth Table of Logic Circuit
- Analyze the Logic Circuit
- Generate Boolean Expression of Logic Circuit
- Classify Types of Logic Circuit
- Analyze the Load Delay of Logic Circuit
- Calculate the Node Centralities of the Logic Circuit Graph
- Compare the structure of 2 logic circuit graphs
- Discover the Feedback Circuit.
- Convert the Logic Graph to Logic Circuit
- Identifying sub-logic circuits (e.g., a decoder) within a larger logic circuit

Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

2

3

Motivation

An AI model needs data to learn. In the field of circuit design, that data is logic circuits. By analyzing thousands or millions of circuit examples, the AI can learn complex design rules and patterns that are difficult for humans to recognize. The main applications include:

- ****Logic Synthesis:**** AI can learn how to convert a Boolean expression or a behavioral description into an optimal circuit diagram.
- ****Optimization:**** AI can suggest changes in an existing circuit to reduce area, lower power consumption, or increase speed (delay reduction).
- ****Performance Prediction:****
- By building a large database with thousands of such entries, you will have a solid foundation for training powerful AI models for logic circuit design.

Prof. Dr. Do Phuc, UIT, VNUHCM, 2025

10/13/2025

3

The main steps for building a database

4

Prof. Dr. Do Phuc, UIT, VNUHCM, 2025

10/13/2025

4

5

Step 1: Convert Logic Circuit to Graph, build a database of Logic Circuit Graph

- This is the most critical step. The data must be in a format that computers and AI algorithms can understand.
- Graph Representation: This is a very powerful and popular method, and the one you used in previous examples. We convert Logic Circuit to Graph
- Nodes: Represent logic gates (AND, OR, NOT, XOR), Inputs, and Outputs.
- Edges: Represent the connections (wires) between the gates.
- Advantage: Perfectly captures the structure and spatial relationships of the circuit. Highly suitable for graph-based AI models like Graph Neural Networks (GNNs).
- Hardware Description Languages (HDL - Verilog/VHDL): Circuits are stored as code. AI can be trained to understand or generate this code, similar to natural language processing AI models.
- Netlist: A low-level text format describing the gates and their connections. This is an intermediate format between HDL and physical layout.
- ➡ Recommendation: Start with the graph representation (nodes/edges) as it is intuitive and highly effective for many modern AI algorithms.

Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

5

6

Step 2: Data Acquisition/Generation

- Use Standard Benchmark Suites: This is the best source for real-world data. Researchers have created benchmark suites to test Electronic Design Automation (EDA) algorithms.
- Examples: ISCAS'85, MCNC, EPFL Benchmark Suite. You can find and download them online.
- Procedural Generation of Circuits: Write scripts to automatically generate random circuits or circuits based on random Boolean expressions.
- Generate a random Boolean expression (e.g., $(A \& B) \vee (\neg C \& D)$).
- Write a program to parse this expression and automatically create the corresponding nodes.csv and edges.csv files.
- Gather from Open Sources: Look for simple circuit designs from textbooks and open-source hardware projects.

Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

6

7

Step 3: Feature Extraction & Labeling

- Structural Features
 - Number of gates of each type (number of AND, OR, NOT gates, etc.).
 - Total number of gates, number of connections.
 - Circuit depth (the longest path from input to output).
- Performance Labels (PPA)
 - These are the most crucial labels for optimization problems.
 - Performance (Speed): The maximum circuit delay (critical path delay).
- Power: Estimated power consumption.
 - Area: The total area occupied on the chip (often estimated by the total number of gates).
- Functional Features
 - The circuit's Truth Table.
 - The corresponding Boolean Expression.
- Note: To obtain these labels, you usually need to use specialized EDA tools (such as Synopsys Design Compiler, Cadence Genus) to synthesize the circuit and generate the reports.

Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

7

8

Step 4: Structuring and Storage

- **Graph Database:** For very large and complex datasets, using a specialized database like Neo4j or TigerGraph is ideal.
- **Compressed File Format:** Store the entire dataset in efficient formats such as HDF5 or Parquet for easy loading into AI frameworks (TensorFlow, PyTorch).

Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

8

9

Challenges

- The precision of converting the Logic Circuit to a Graph
- The precision of Logic Graph Analysis
- Large Logic Circuit Processing, Analysis
-
- We are in a running process!

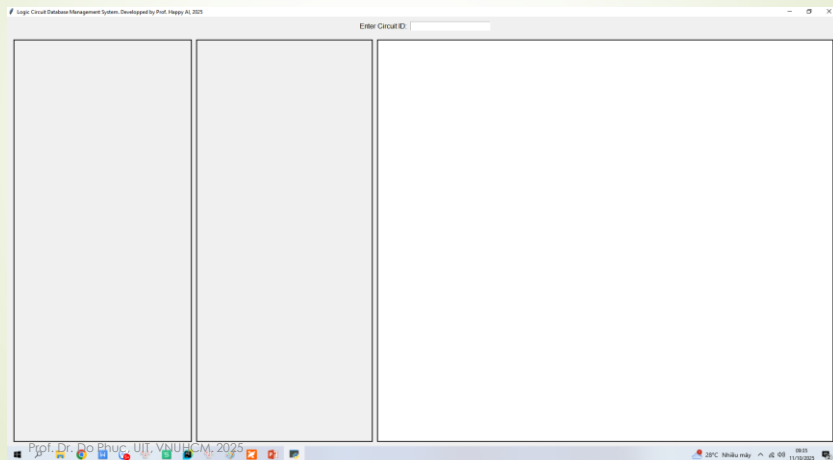
Prof. Dr. Do Phuc, UIT, VNUHCM, 2025

10/13/2025

9

10

Interface of our Logic Circuit Database



10/13/2025

10

[illegible]

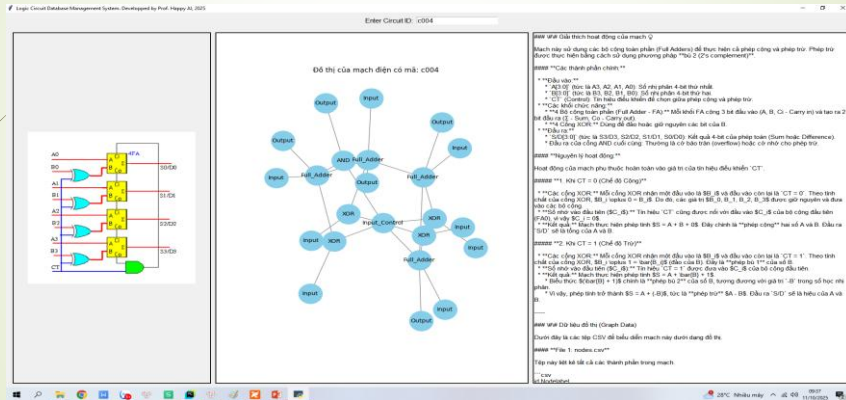
10/13/2025

[illegible]

10/13/2025

13

Some results of Our Logic Circuit Database for Circuit Retrieval



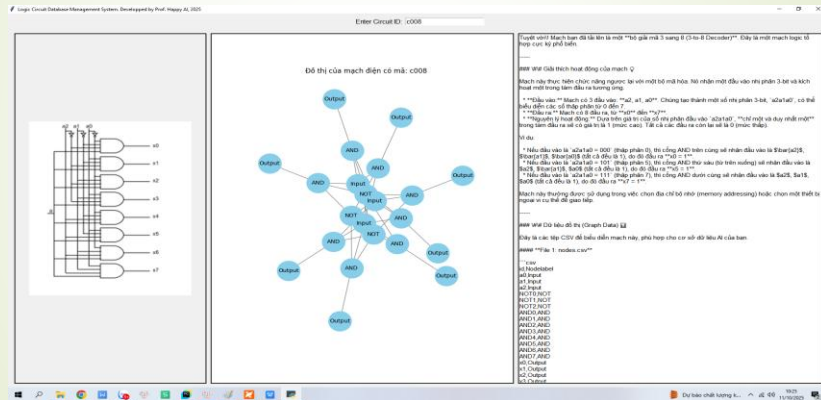
Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

13

14

Some results of Our Logic Circuit Database for Circuit Retrieval



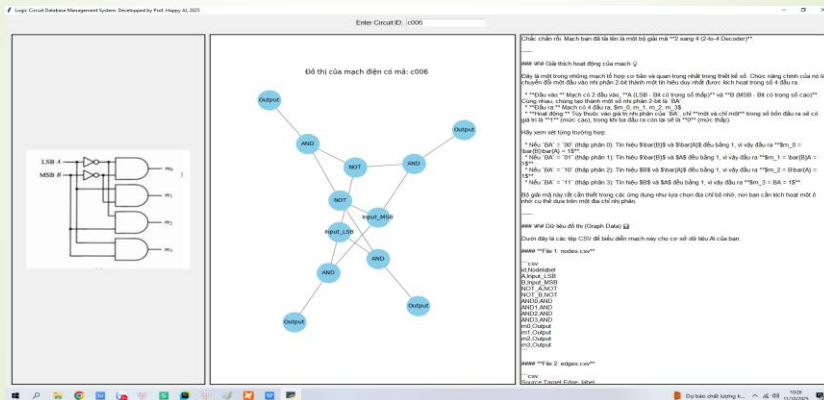
Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

14

15

Some results of Our Logic Circuit Database for Circuit Retrieval



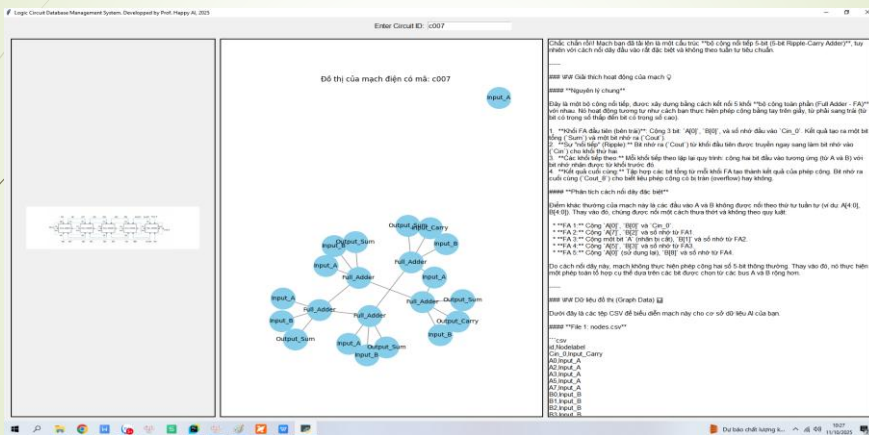
Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

15

16

Some results of Our Logic Circuit Database for Circuit Retrieval



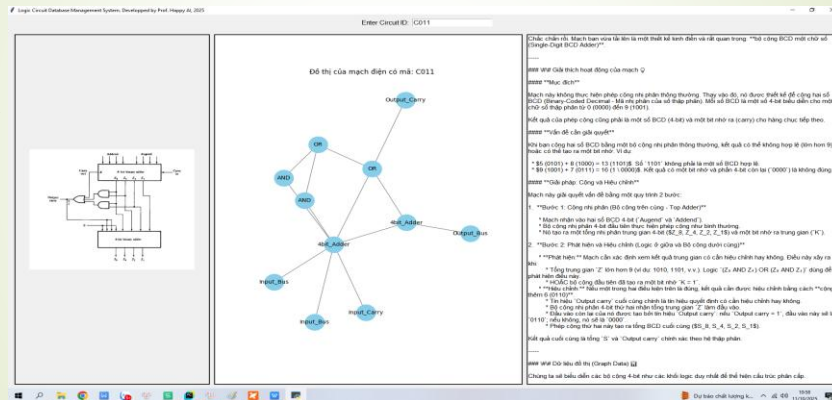
Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

16

19

Some results of Our Logic Circuit Database for Circuit Retrieval



Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

19

Logic Circuit Analysis

20

Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

20

Build True Table of Logic Circuit

21

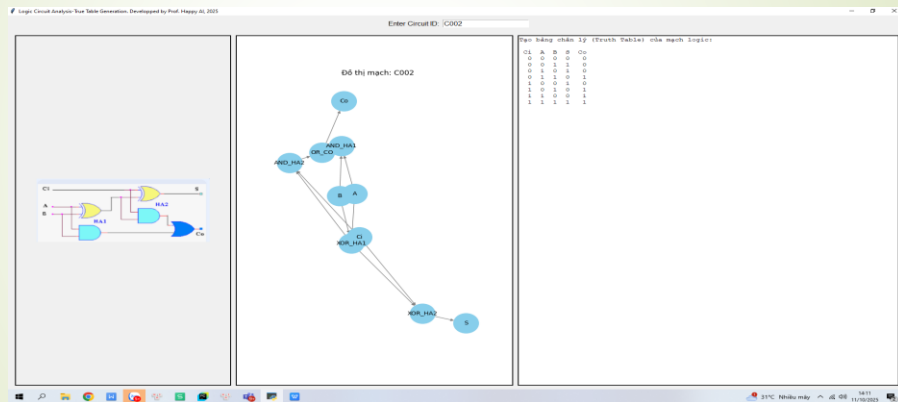
Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

21

22

Some results of buiding True Table of Logic Circuit



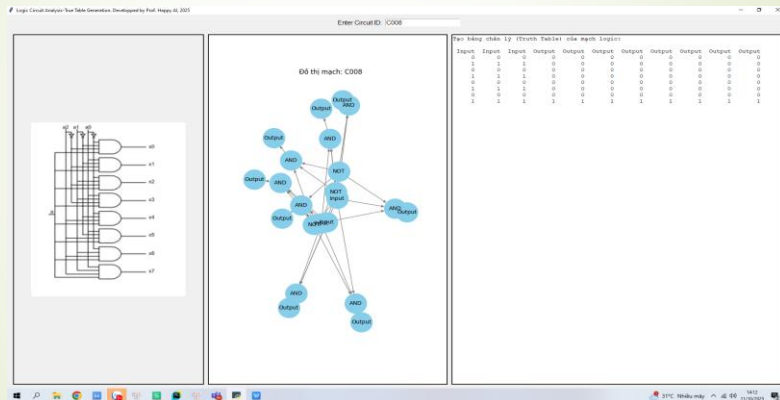
Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

22

23

Some results of buiding True Table of Logic Circuit



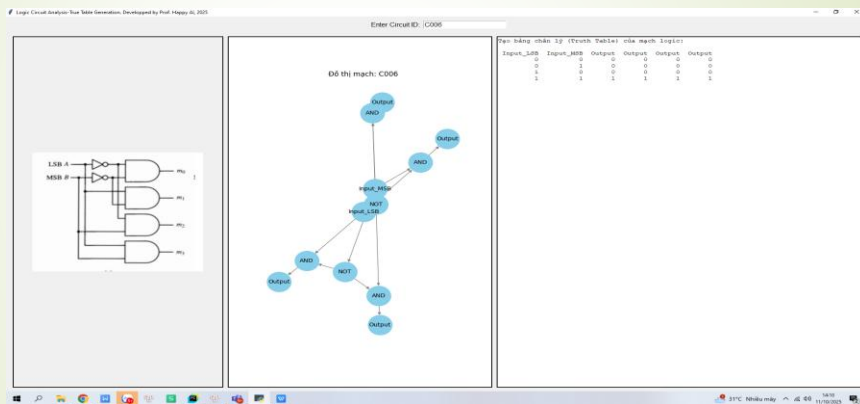
Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

23

24

Some results of buiding True Table of Logic Circuit



Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

24

Count Gate Type Of Logic Circuit

25

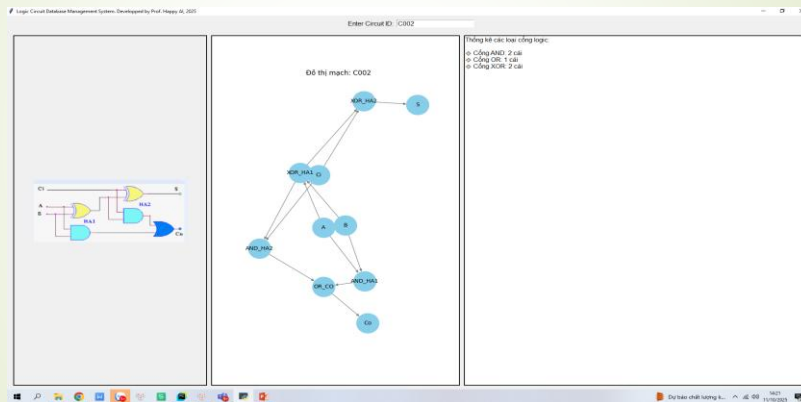
Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

25

26

Some Results of Counting The Gates of Logic Circuit



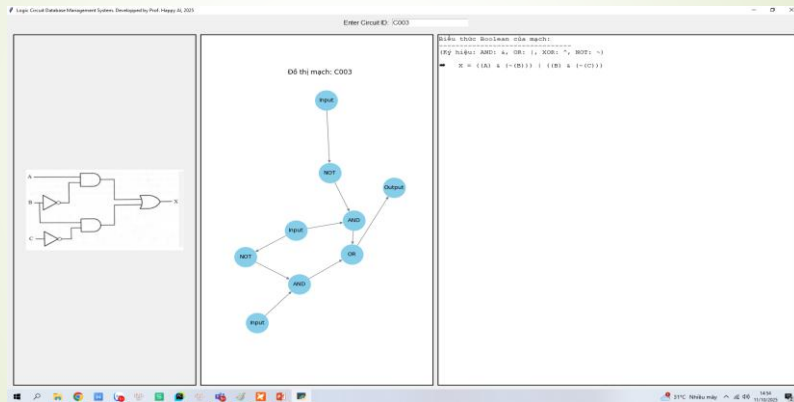
Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

26

31

Some Results of Generating Boolean Expression of Logic Circuit



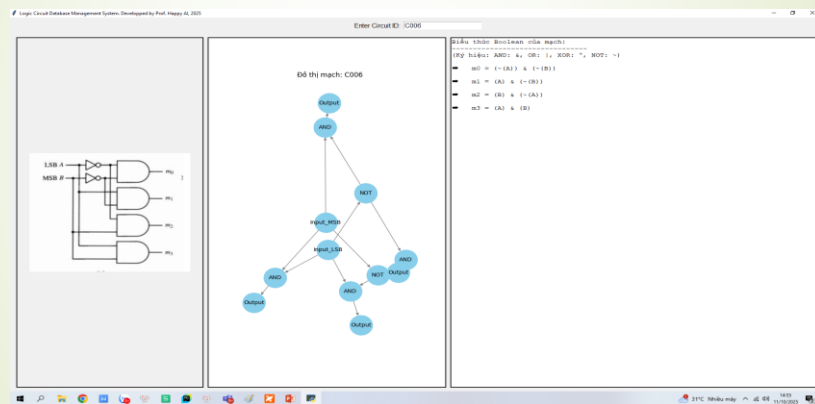
Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

31

32

Some Results of Generating Boolean Expression of Logic Circuit



Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

32

Classify Type of Logic Circuit

33

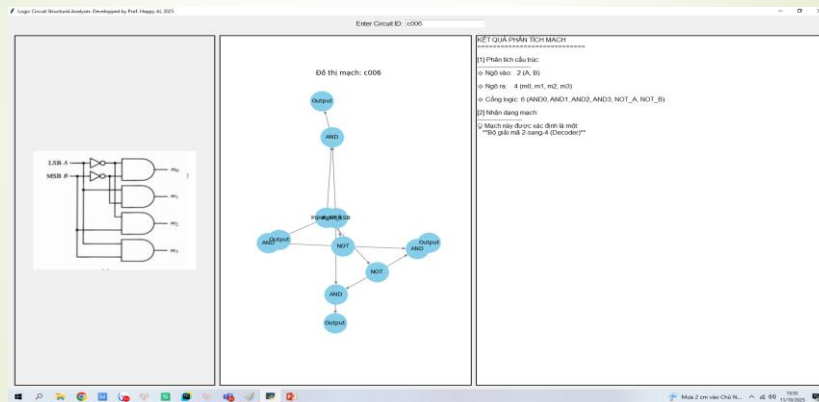
Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

33

34

Some Results of Classifying Type of Logic Circuit



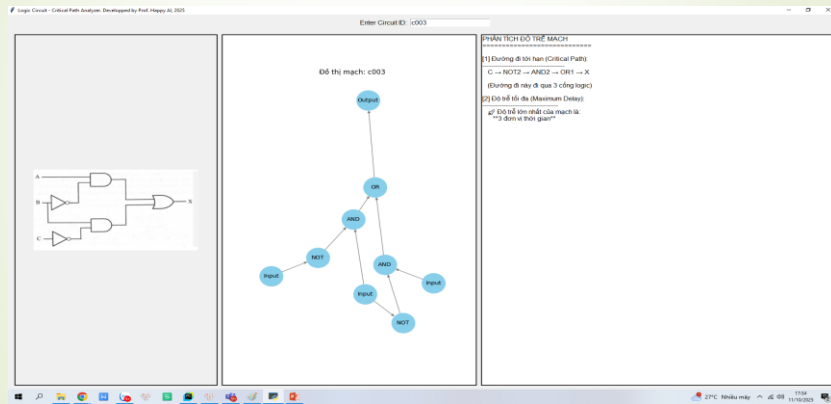
Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

34

39

Some Results of Load Delay Analysis of Logic Circuit



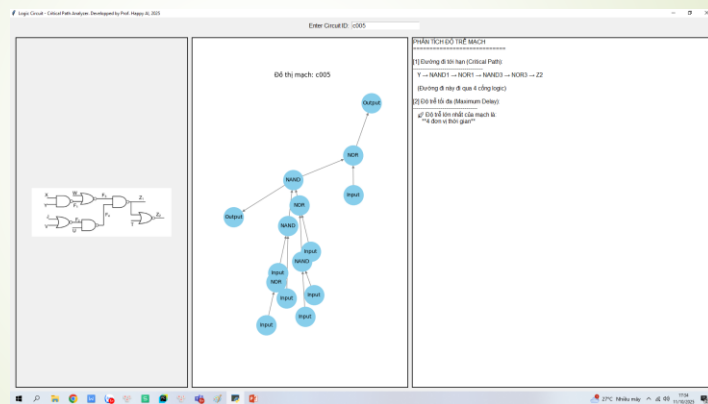
Prof. Dr. Do Phuc, UIT, VNUHCM, 2025

10/13/2025

39

40

Some Results of Load Delay Analysis of Logic Circuit



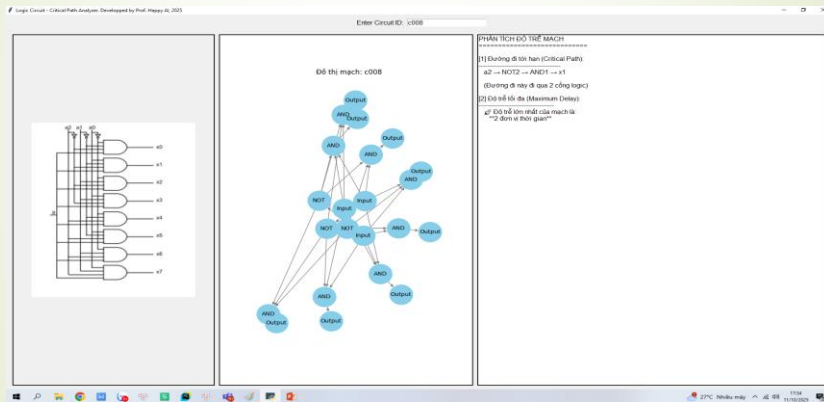
Prof. Dr. Do Phuc, UIT, VNUHCM, 2025

10/13/2025

40

41

Some Results of Load Delay Analysis of Logic Circuit



Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

41

Calculate the Node Centralities of the Logic Circuit Graph

42

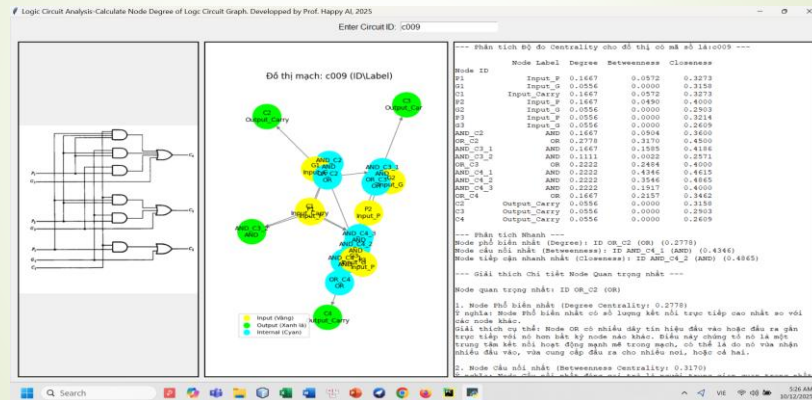
Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

42

45

Node Centrality Analysis of Logic Circuit Graph



Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

45

Compare the structure of 2 Logic circuit graphs

46

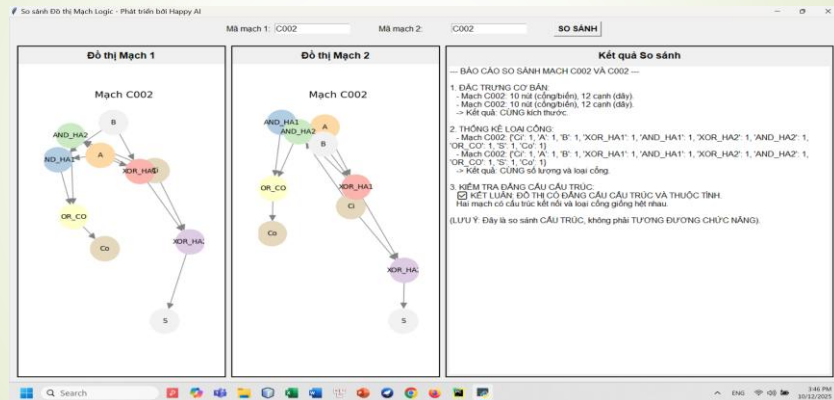
Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

46

47

Compare the structure of 2 logic circuit graphs



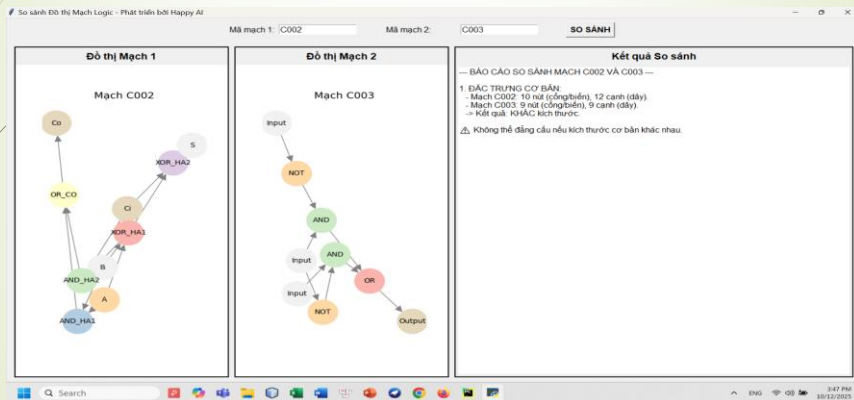
Prof. Dr. Do Phuc, UIT, VNUHCM, 2025

10/13/2025

47

48

Compare the structure of 2 logic circuit graphs



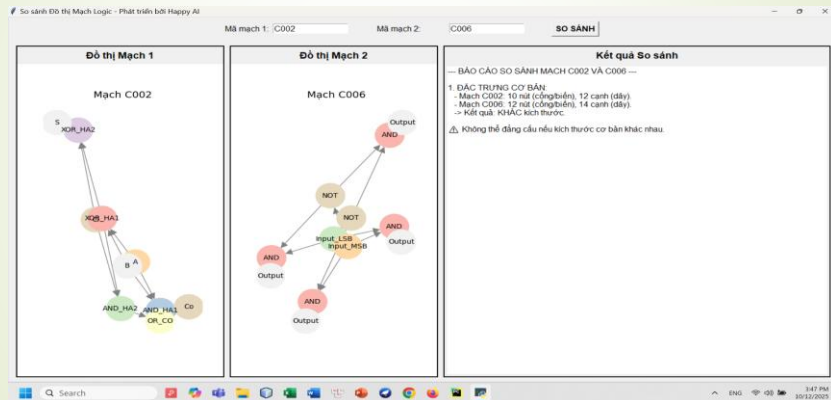
Prof. Dr. Do Phuc, UIT, VNUHCM, 2025

10/13/2025

48

49

Compare the structure of 2 logic circuit graphs



Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

49

Discover the Feedback Circuit

50

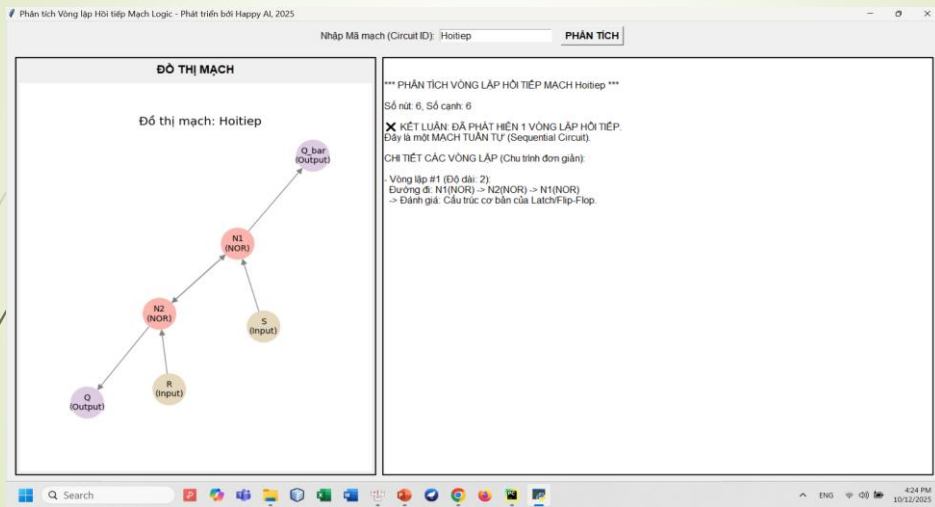
Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

50

51

Discover the feedback circuit



Prof. Dr. Do Phuc, UIT, VNUHCM, 2025

10/13/2025

51

Covert the Logic Graph to Logic Circuit

52

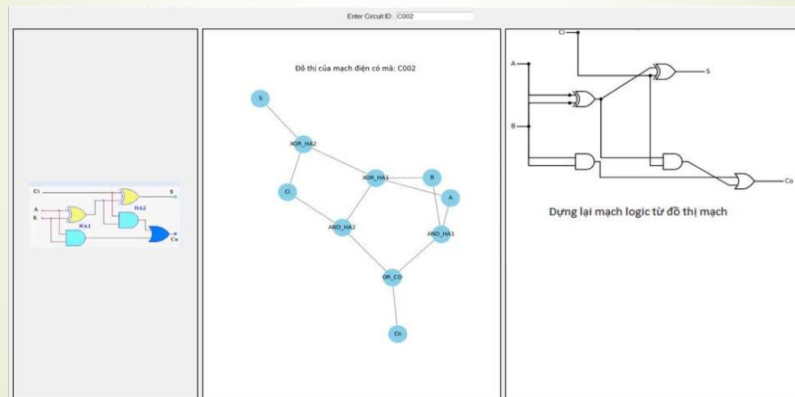
Prof. Dr. Do Phuc, UIT, VNUHCM, 2025

10/13/2025

52

53

Covert the Logic Graph to Logic Circuit



Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

53

Identifying sub-logic circuits (e.g., a **decoder**) within a larger logic circuit

54

Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

54

55

Extremely important in **design, analysis, debugging, and optimization** of electronic circuits.

- **1. Analysis and Understanding of Circuit Structure**
 - **System Decomposition:** Allows a complex circuit to be broken down into smaller, more manageable, and easier-to-understand functional blocks. For example, recognizing a decoder indicates that a part of the circuit is performing **address decoding** or **device/channel selection**.
 - **Function Identification:** Helps quickly determine the **specific function** that the sub-circuit is performing within the larger system. This is the first step in understanding how the entire circuit operates.
- **2. Modular Design and Reusability**
 - **Bottom-up design:** By using standardized sub-logic blocks (like decoders, multiplexers, registers, counters), designers can build larger circuits in a **structured** and **efficient** manner.
 - **Reusability:** Verified and tested sub-circuits can be reused in multiple different projects, which **saves design time** and **minimizes errors**.
 - **Modularization:** Encourages modular design, making it easier to **replace** or **upgrade** a specific part of the circuit.

Prof. Dr. Do Phuc, UIT, VNUHCM, 2025

10/13/2025

55

56

Extremely important in **design, analysis, debugging, and optimization** of electronic circuits.

- **3. Debugging and Testing**
 - **Fault Isolation:** If the larger circuit is not functioning correctly, identifying the sub-circuits allows for **isolating** the problem. Instead of checking the entire circuit, the engineer only needs to focus on the inputs/outputs of the suspected sub-circuit.
 - **Unit Testing:** Sub-circuits can be tested independently (**unit testing**), ensuring that each basic element works correctly before integrating them.
- **4. Circuit Optimization**
 - **Performance Improvement:** Recognizing a sub-circuit (like a decoder) may lead to replacing it with a more **optimized** version (e.g., faster, lower power consumption) without needing to change the rest of the circuit.
 - **Simplification:** In analysis, if a group of logic gates performs the function of a standard sub-circuit, it can sometimes **simplify** the representation of the circuit diagram or its Boolean equations.

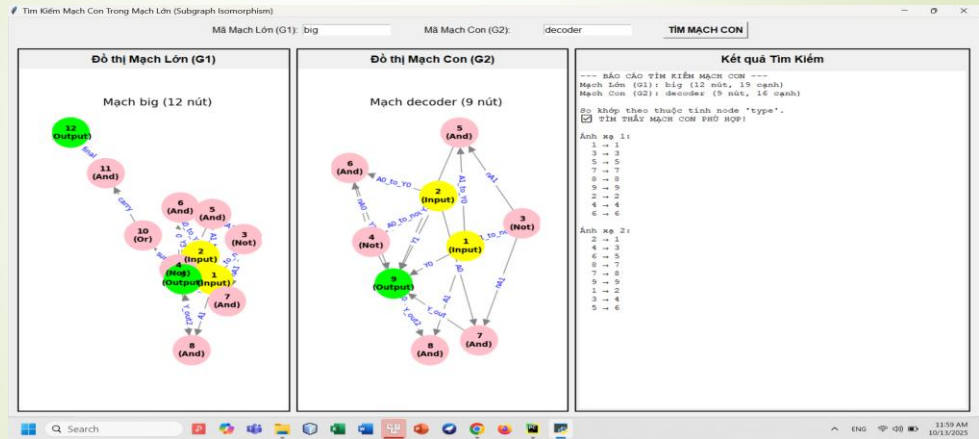
Prof. Dr. Do Phuc, UIT, VNUHCM, 2025

10/13/2025

56

57

Example of Identifying sub-logic circuits (e.g., a **decoder**) within a larger logic circuit



Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

57

Thank you for your attention!

58

Please contact: Prof. Dr. Do Phuc
 Email: phucdo@uit.edu.vn
 Year 2025

Prof. Dr. Do Phuc, UIT, VNUHCM. 2025

10/13/2025

58